

DIABETES RISK & HEALTH ANALYTICS

End-to-End Data Analytics Project

An end-to-end analytics project focused on identifying diabetes prevalence patterns, health and lifestyle risk factors, and high-risk population segments using data cleaning, exploratory analysis, SQL analytics, DAX, and interactive Power BI dashboards.

Tools Used:

Python | PostgreSQL | Power BI | DAX

Project Date:

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1. Project Overview

Project Summary:

Diabetes Risk & Health Analytics is an end-to-end data analytics project focused on analyzing health, lifestyle, demographic, and clinical factors associated with diabetes prevalence.

The project uses a dataset containing **100,000 records and 31 variables** covering demographic characteristics, lifestyle behaviors, medical history, body measurements, cardiovascular indicators, biomarkers, diabetes risk scores, and diagnosed diabetes status.

The analysis follows a structured workflow using **Python, PostgreSQL, and Power BI**. Python was used for data understanding, data cleaning, validation, and exploratory data analysis. PostgreSQL was used to perform business-focused SQL analysis, calculate key performance indicators, identify patterns across population segments, and conduct advanced risk analysis. Power BI was then used to build an interactive dashboard that presents the key findings in a clear and business-oriented format.

What this Project Analyzes:

- Diabetes prevalence across demographic groups
- Lifestyle factors such as physical activity, diet, smoking, alcohol consumption, sleep, and screen
- Medical history including family history, hypertension, and cardiovascular history
- BMI and waist-to-hip ratio
- Blood pressure and cardiovascular indicators
- Glucose, HbA1c, insulin, and lipid biomarkers
- Diabetes risk-score distribution and risk quartiles
- High-risk individuals who are currently undiagnosed
- Relationships between health factors and diabetes prevalence

Analytical Approach:

Python → PostgreSQL → Power BI → Insights & Recommendations

Project Outcome:

The final deliverable is a four-page Power BI dashboard covering Executive Overview, Risk Factors, Health & Biomarkers, and High-Risk Population.

2. Business Objective

Objective:

The primary objective of this project is to analyze diabetes-related health and lifestyle data to identify patterns associated with diabetes prevalence, understand major risk factors, and identify population segments that may require greater attention.

The analysis is designed to transform raw healthcare data into **clear, measurable, and actionable business insights** using Python, SQL, and Power BI.

Key Business Questions:

1. Diabetes Prevalence

- What is the overall diabetes prevalence in the dataset?
- How does diabetes prevalence vary across different age groups and demographic segments?

2. Lifestyle & Behavioral Factors

- How does diabetes prevalence vary by physical activity and diet quality?
- What patterns are observed across smoking, alcohol consumption, sleep, and screen time?

3. Medical & Physical Risk Factors

- How is diabetes prevalence associated with family history, hypertension, and cardiovascular history?
- How does BMI relate to diabetes prevalence?

4. Health & Biomarker Analysis

- How does diabetes prevalence vary across fasting glucose and HbA1c categories?
- What patterns are observed across cholesterol, triglycerides, and blood pressure categories?

5. Risk Segmentation

- How does diabetes prevalence change across diabetes risk-score quartiles?
- How many high-risk individuals are currently undiagnosed?

Analytical Goals:

- Measure diabetes prevalence and key health indicators.
- Compare major lifestyle and health-related risk factors.
- Segment individuals based on diabetes risk scores.
- Identify high-risk individuals who are currently undiagnosed.
- Translate analytical findings into practical recommendations.

3. Dataset Overview

Dataset Description:

The dataset contains **100,000 records and 31 variables** related to diabetes risk, health characteristics, lifestyle behaviors, demographic information, and clinical biomarkers.

Each row represents an individual, while the columns capture demographic attributes, lifestyle factors, medical history, physical measurements, health indicators, diabetes risk scores, and diabetes outcomes.

Dataset Dimensions:

Attribute	Details
Total Records	100,000
Total Variables	31
Data Type	Structured tabular data
Domain	Healthcare / Diabetes Risk
Target Variable	diagnosed_diabetes
Target Values	0 = No, 1 = Yes
Primary Analysis	Diabetes Prevalence & Risk Factors

Variable Categories:

Category	Key Variables
Demographics	Age, Gender, Ethnicity, Education Level, Income Level, Employment Status
Lifestyle	Smoking Status, Alcohol Consumption, Physical Activity, Diet Score, Sleep, Screen Time
Medical History	Family History, Hypertension History, Cardiovascular History
Body Measurements	BMI, Waist-to-Hip Ratio
Vital Signs	Systolic BP, Diastolic BP, Heart Rate
Lipid Profile	Total Cholesterol, HDL, LDL, Triglycerides
Glucose & Biomarkers	Fasting Glucose, Postprandial Glucose, Insulin, HbA1c
Risk & Outcome	Diabetes Risk Score, Diabetes Stage, Diagnosed Diabetes

Target Variable:

The primary outcome variable is **diagnosed_diabetes**.

- 0 → Not diagnosed with diabetes
- 1 → Diagnosed with diabetes

4. Project Workflow

Workflow Overview:

The project follows an end-to-end data analytics workflow, beginning with raw healthcare data and progressing through data preparation, exploratory analysis, SQL-based analysis, visualization, and business recommendations.

1. Data Understanding & Preparation — Python:

- Data understanding
- Cleaning & validation
- Missing values, duplicates & outliers

2. Exploratory Analysis — Python:

- Demographics
- Lifestyle & medical factors
- Health & biomarker analysis

3. Business Analysis — PostgreSQL:

- KPIs & business questions
- Risk-factor analysis
- High-risk population segmentation

4. Data Modeling — Power BI:

- Data model
- DAX measures and Calculated Columns

5. Dashboard Development – Power BI:

- Executive Overview
- Risk Factors
- Health & Biomarkers
- High-Risk Population

6. Insights & Recommendations:

- Key findings and Risk Factors
- Data-driven recommendations



5. Python — Data Cleaning & EDA

Objective:

Python was used to understand, clean, validate, and explore the healthcare dataset before performing SQL analysis and dashboard development.

5.1 Data Understanding:

The dataset was initially examined to understand its structure, dimensions, variables, data types, and overall characteristics. The dataset contains **100,000 records and 31 variables** covering demographic, lifestyle, medical, health, and diabetes-related information.

include:

- head()
- sample()
- shape()
- columns()
- dtypes()
- info()

```
[5]: # Check the shape of the data
df_clean.shape

[5]: (100000, 31)

[6]: # check the sample rows
df_clean.sample(5)

[6]:
```

	age	gender	ethnicity	education_level	income_level	employment_status	smoking_status	alcohol_consumption_per_week	physical_activity_r
66692	56	Male	Black	No formal	Upper-Middle	Retired	Never	3	
58769	23	Female	White	Graduate	Upper-Middle	Employed	Never	2	
42410	58	Female	White	Graduate	Middle	Employed	Never	1	
13877	50	Male	White	Highschool	Lower-Middle	Employed	Current	2	
94685	69	Male	Hispanic	Postgraduate	Lower-Middle	Unemployed	Current	1	

5.2 Data Quality checks:

Data quality checks were performed to assess the reliability and consistency of the dataset before analysis. The checks focused on missing values, duplicate records, categorical consistency, data types, descriptive statistics, and potential outliers.

5.3 Missing Values:

Missing-value analysis was performed across all variables to identify incomplete records. The results were reviewed to determine whether any data-cleaning action was required before further analysis.

```
[10]: # Check the missing values in the data
      df_clean.isnull().sum()

[10]: age                0
      gender              0
      ethnicity           0
      education_level     0
      income_level        0
      employment_status   0
      smoking_status       0
      alcohol_consumption_per_week  0
      physical_activity_minutes_per_week  0
      diet_score           0
      sleep_hours_per_day  0
      screen_time_hours_per_day  0
      family_history_diabetes  0
      hypertension_history  0
      cardiovascular_history  0
      bmi                 0
      waist_to_hip_ratio   0
      systolic_bp          0
      diastolic_bp         0
      heart_rate           0
      cholesterol_total    0
      hdl_cholesterol      0
      ldl_cholesterol      0
      triglycerides        0
      glucose_fasting      0
      glucose_postprandial  0
      insulin_level        0
      hba1c               0
      diabetes_risk_score  0
      diabetes_stage       0
      diagnosed_diabetes   0
      dtype: int64
```

5.4 Duplicate Check:

Duplicate records were checked to ensure that the dataset did not contain unintended repeated observations.

```
[11]: # Check the duplicate rows in the data
      df_clean.duplicated().sum()

[11]: np.int64(0)
```

5.5 Outlier Analysis:

The IQR method was used to identify statistically extreme observations in numerical variables. The flagged values were reviewed against their observed ranges and domain context. Plausible extreme health values were retained rather than automatically removed.

```

15]: # Check outliers in the data
for cols in numeric_cols:
    Q1 = df_clean[cols].quantile(0.25)
    Q3 = df_clean[cols].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df_clean[(df_clean[cols] < lower_bound) | (df_clean[cols] > upper_bound)]
    print(f"{cols}: {len(outliers)} outliers")

age: 0 outliers
alcohol_consumption_per_week: 458 outliers
physical_activity_minutes_per_week: 3199 outliers
diet_score: 337 outliers
sleep_hours_per_day: 900 outliers
screen_time_hours_per_day: 305 outliers
family_history_diabetes: 21941 outliers
hypertension_history: 0 outliers
cardiovascular_history: 7920 outliers
bmi: 744 outliers
waist_to_hip_ratio: 273 outliers
systolic_bp: 530 outliers
diastolic_bp: 731 outliers
heart_rate: 855 outliers
cholesterol_total: 309 outliers
hdl_cholesterol: 565 outliers
ldl_cholesterol: 349 outliers
triglycerides: 301 outliers
glucose_fasting: 745 outliers
glucose_postprandial: 634 outliers
insulin_level: 326 outliers
hba1c: 618 outliers
diabetes_risk_score: 914 outliers
diagnosed_diabetes: 0 outliers

```

5.6 Exploratory Data Analysis:

Exploratory Data Analysis was performed to understand the distribution of key variables and identify patterns and relationships related to diabetes prevalence.

Analysis Performed:

The EDA was conducted across the following areas:

- **Demographics:** Age, gender, ethnicity, education, income, and employment
- **Lifestyle:** Smoking, alcohol consumption, physical activity, diet, sleep, and screen time
- **Medical History:** Family history, hypertension, and cardiovascular history
- **Body & Vitals:** BMI, waist-to-hip ratio, blood pressure, and heart rate
- **Biomarkers:** Cholesterol, triglycerides, glucose, insulin, and HbA1c
- **Diabetes Outcomes:** Risk score, diabetes stage, and diagnosed diabetes

Key EDA Finding:

These findings were used to guide the subsequent **SQL business analysis and Power BI dashboard development.**

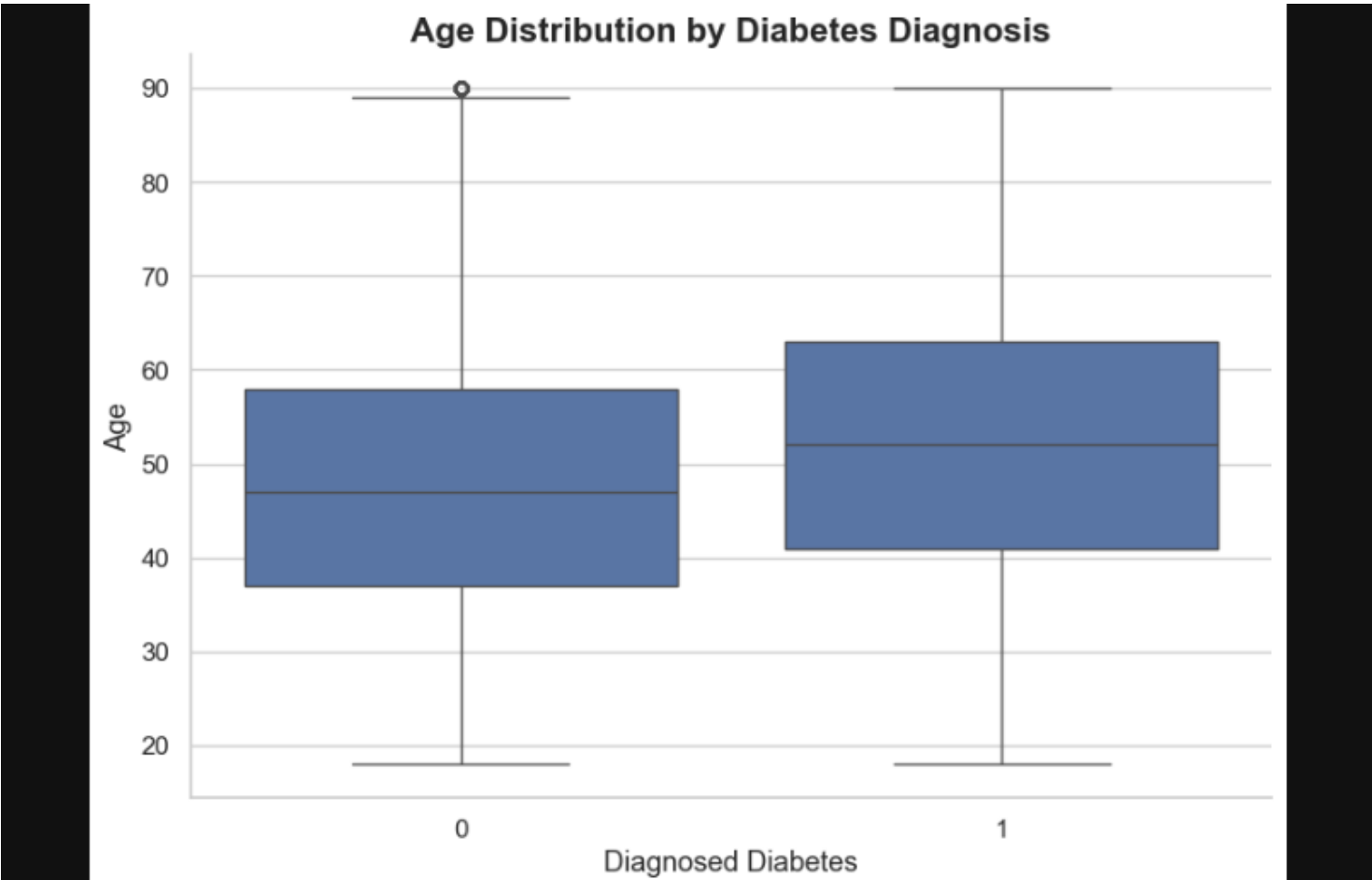


Figure 1. Age Distribution by Diabetes Diagnosis

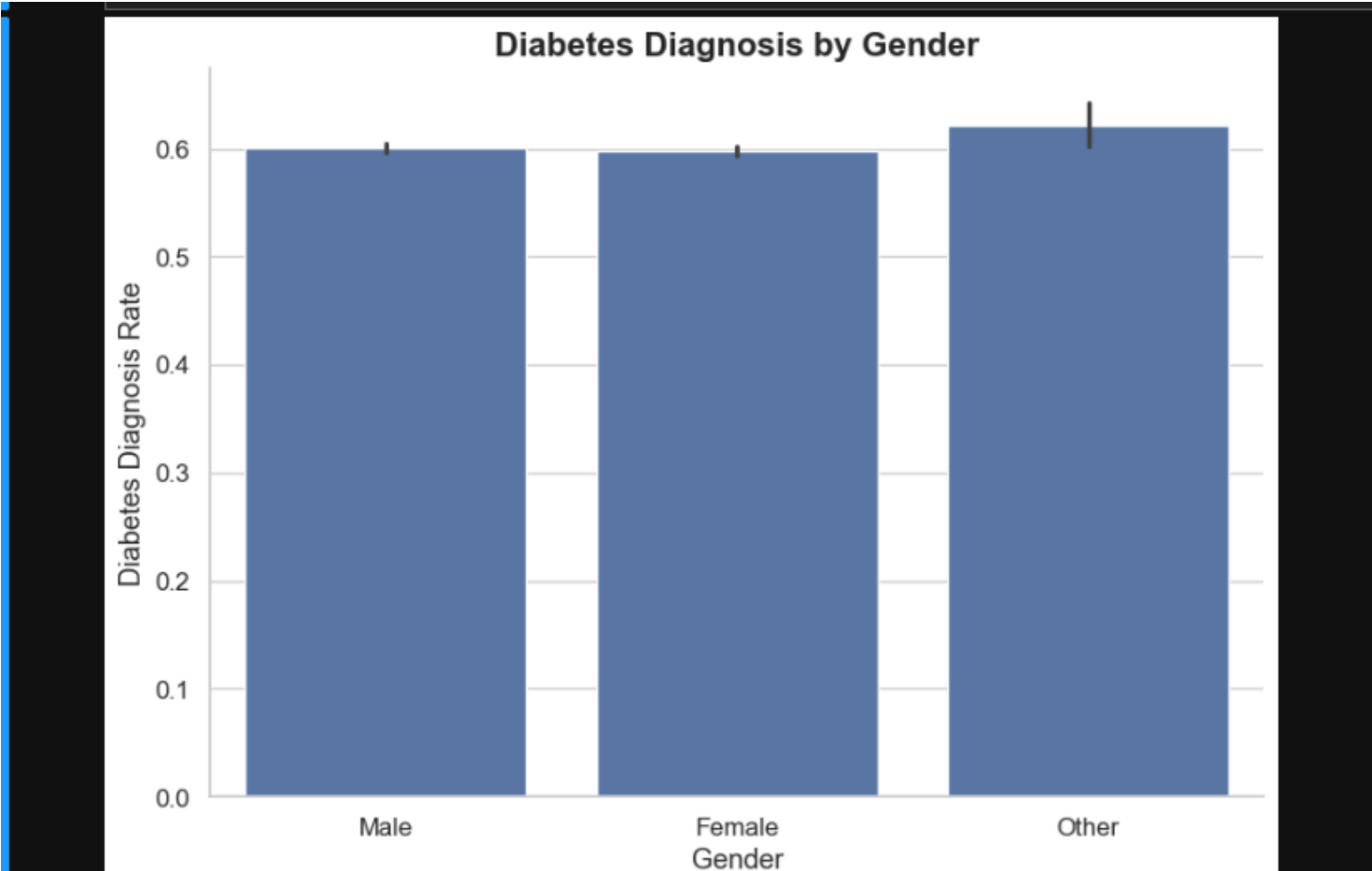


Figure 2. Diabetes Prevalence by Gender

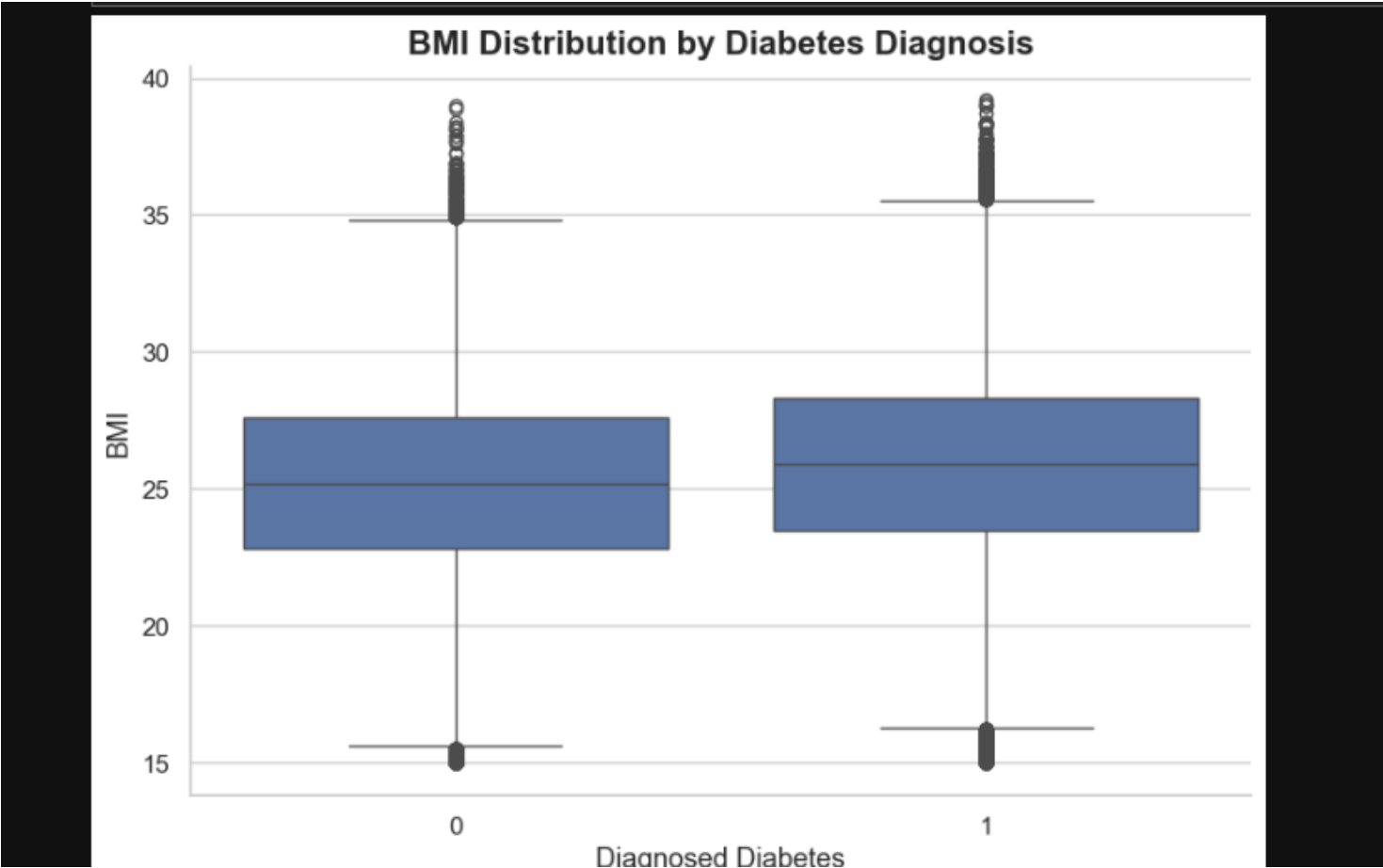


Figure 3. BMI Distribution by Diabetes Diagnosis

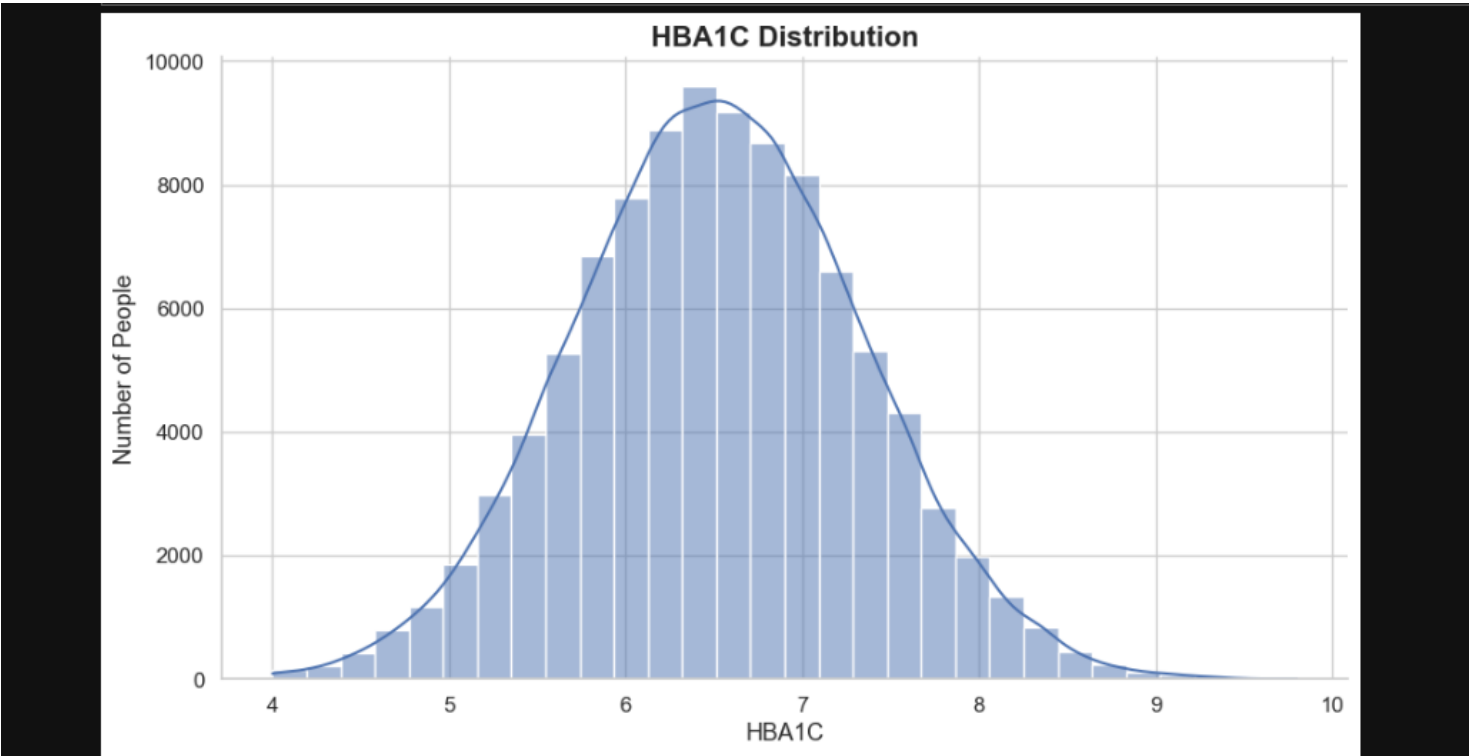


Figure 4. HbA1c Distribution

6. SQL Analysis

Overview:

PostgreSQL was used to perform business-focused analysis on the cleaned dataset. SQL queries were developed to calculate key metrics, compare population segments, analyze diabetes-related risk factors, and identify high-risk individuals.

6.1 Import Data:

Clean data was imported from the csv file which we created in the jupyter notebook. For this we have created database inside the pgAdmin and created table for this data.

6.2 Data Overview:

- Display 10 rows
- Check total number of rows
- Check null values
- Check duplicate rows

6.3 KPI Analysis:

KPI	Result
Total People	100,000
Diabetes Cases	59,998
Non-Diabetes Cases	40,002
Diabetes Prevalence	60.00%
Average Risk Score	30.22
Average BMI	25.61
Average HbA1c	6.52
Average Fasting Glucose	111.12

6.4 Demographic Analysis:

Diabetes prevalence was compared across:

- Age groups
- Gender
- Ethnicity
- Education level
- Income level

- Employment status

Age-Group vs Diabetes:

```
WITH age_group_cte AS(
SELECT
diagnosed_diabetes,
CASE
WHEN age < 30 THEN 'Under 30'
WHEN age < 45 THEN '30-44'
WHEN age < 60 THEN '45-59'
ELSE '60+'
END AS age_group
FROM diabetes_risk
)
SELECT
age_group,
COUNT(*) AS total_people,
SUM(
CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END
) AS diabetes_cases,
ROUND(
SUM(CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END) * 100.0
/ COUNT(*),
2
) AS diabetes_rate_pct
FROM age_group_cte
GROUP BY age_group
ORDER BY diabetes_rate_pct DESC;
```

Data Output					Messages	Notifications
	age_group text	total_people bigint	diabetes_cases bigint	diabetes_rate_pct numeric		
1	60+	27615	18879	68.37		
2	45-59	35814	21734	60.69		
3	30-44	26569	14599	54.95		
4	Under 30	10002	4786	47.85		

Key finding: Diabetes prevalence was higher among older age groups, with the **60+ group showing 68.37% prevalence**.

6.5 Lifestyle Analysis:

Lifestyle factors were analyzed to identify differences in diabetes prevalence across:

- Physical activity
- Diet quality
- Smoking
- Alcohol consumption
- Sleep
- Screen time

Physical Activity vs Diabetes:

```
WITH activity_level_cte AS(
SELECT
diagnosed_diabetes,
CASE
WHEN physical_activity_minutes_per_week < 150 THEN 'Low'
WHEN physical_activity_minutes_per_week < 300 THEN 'Medium'
ELSE 'High'
END AS activity_level
FROM diabetes_risk
)
SELECT
activity_level,
COUNT(*) AS total_people,
SUM(
CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END
) AS diabetes_cases,
ROUND(
SUM(CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END) * 100.0
/ COUNT(*),
2
) AS diabetes_rate_pct
FROM activity_level_cte
GROUP BY activity_level
ORDER BY diabetes_rate_pct DESC;
```

Data Output

Messages

Notifications

	activity_level text	total_people bigint	diabetes_cases bigint	diabetes_rate_pct numeric
1	Low	71505	44762	62.60
2	Medium	24531	13392	54.59
3	High	3964	1844	46.52

Key finding: Diabetes prevalence was **62.60% among the low physical-activity group compared with 46.52% among the high-activity group.**

6.6 Medical & Health Analysis:

SQL was used to analyze relationships between diabetes prevalence and:

- Family history
- Hypertension
- Cardiovascular history
- BMI
- Blood pressure
- Glucose
- HbA1c
- Cholesterol and triglyceride

Family History vs Diabetes:

```
SELECT
family_history_diabetes,
COUNT(*) AS total_people,
SUM(
CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END
) AS diabetes_cases,
ROUND(
SUM(CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END) * 100.0
/ COUNT(*),
2
) AS diabetes_rate_pct
```

```
FROM diabetes_risk
GROUP BY family_history_diabetes
ORDER BY diabetes_rate_pct DESC;
```

Data Output

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SQL

Show

	family_history_diabetes integer	total_people bigint	diabetes_cases bigint	diabetes_rate_pct numeric
1	1	21941	17177	78.29
2	0	78059	42821	54.86

Key finding: Individuals with a family history of diabetes showed **78.29% prevalence compared with 54.86% among those without a recorded family history.**

6.7 Biomarkers:

SQL was used to analyze relationships between diabetes prevalence and:

- Fasting Glucose
- Postprandial Glucose
- HbA1c
- Insulin Level

Fasting Glucose — 60 to 172 mg/dL

```
WITH fasting_glucose_group_cte AS(
SELECT
diagnosed_diabetes,
CASE
    WHEN glucose_fasting < 70 THEN 'Low'
    WHEN glucose_fasting < 100 THEN 'Normal'
    WHEN glucose_fasting < 126 THEN 'Prediabetes Range'
    ELSE 'Diabetes Range'
END AS fasting_glucose_group
FROM diabetes_risk
)
```

```
SELECT
```

```
COUNT(*) AS total_people,
SUM(
    CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END
) AS diabetes_cases,
ROUND(
    SUM(CASE WHEN diagnosed_diabetes = 1 THEN 1 ELSE 0 END) * 100.0
    / COUNT(*),
    2
) AS diabetes_rate_pct
FROM fasting_glucose_group_cte
GROUP BY fasting_glucose_group
ORDER BY diabetes_rate_pct DESC;
```

Data OutputMessagesNotifications

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SQL

Show

	fasting_glucose_group text	total_people bigint	diabetes_cases bigint	diabetes_rate_pct numeric
1	Diabetes Range	14528	14528	100.00
2	Prediabetes Range	65757	40675	61.86
3	Normal	19618	4784	24.39
4	Low	97	11	11.34

Key finding: Diabetes prevalence increased substantially across glucose categories, ranging from **11.34% in the low group to 100% in the diabetes-range group**.

6.8 High-Risk People:

A high-risk segment was defined using:

Diabetes Risk Score ≥ 40 and Diagnosed Diabetes = 0

The analysis identified **2,856 high-risk, currently undiagnosed individuals**, representing **2.86% of the total population**.

Biggest high-risk segments

```
WITH high_risk AS (  
  SELECT  
    CASE  
      WHEN age < 30 THEN 'Under 30'  
      WHEN age < 45 THEN '30-44'  
      WHEN age < 60 THEN '45-59'  
      ELSE '60+'  
    END AS age_group,  
    diagnosed_diabetes  
  FROM diabetes_risk  
  WHERE diabetes_risk_score >= 40  
    AND diagnosed_diabetes = 0  
)  
SELECT  
  age_group,  
  COUNT(*) AS high_risk_people  
FROM high_risk  
GROUP BY age_group  
ORDER BY high_risk_people DESC;
```

Data OutputMessagesNotifications

	age_group text	high_risk_people bigint
1	45-59	1187
2	60+	1063
3	30-44	545
4	Under 30	61

The largest high-risk age segment is **45–59 years**, with **1,187 individuals**.

6.9 Advanced Analysis:

Advanced SQL techniques were used to perform deeper segmentation and comparison, including:

- Risk-score quartiles
- Multiple-risk-factor analysis
- Age × BMI segmentation
- Risk-factor comparisons
- Correlation analysis

Key Finding:

Risk Quartile	Diabetes Prevalence
Q1 – Lowest Risk	43.21%
Q2 – Lowest-Mid Risk	55.08%
Q3 – Upper-Mid Risk	63.32%
Q4 – Highest Risk	78.69%

The difference between Q1 and Q4 was **35.48 percentage points**, supporting the usefulness of risk-score segmentation within this dataset.

7. Power BI Dashboard

Overview:

Power BI was used to transform the analyzed dataset into an interactive dashboard for monitoring diabetes prevalence, risk factors, health indicators, and high-risk population segments.

The dashboard combines **data modeling, DAX measures, calculated columns, filters, and interactive visualizations** to present the analysis in a clear and business-oriented format.

7.1 Data Model:

The prepared dataset was imported into Power BI and organized for analytical reporting.

Key activities included:

- Data type verification
- Field organization
- Creation of calculated columns
- Development of analytical groupings
- Preparation of fields for interactive filtering

7.2 DAX & KPI Development:

- Total People
- Diabetes Cases
- Non-Diabetes Cases
- Diabetes Prevalence
- Average Risk Score
- Average Physical Activity
- Obesity %
- Family History %
- Hypertension %
- Average Diet Score
- Average BMI
- Average HbA1c
- Average Fasting Glucose
- Average Postprandial Glucose
- Average Total Cholesterol

- Average Systolic BP
- High-Risk Undiagnosed Population
- High-Risk Population %
- Multiple-Risk Factor Population
- Largest Average High-Risk Age Segment

We have renamed the table name in Power BI from **diabetes_risk** to **DiabetesData** for readability

Diabetes Prevalence % =

```
DIVIDE(  
    [Diabetes Cases],  
    [Total People],  
    0  
)
```

High-Risk Undiagnosed =

```
CALCULATE(  
    COUNTROWS(DiabetesData),  
    DiabetesData[diabetes_risk_score] >= 40,  
    DiabetesData[diagnosed_diabetes] = 0  
)
```

7.3 Calculated Columns:

Calculated columns were also created for analytical segmentation, including **Age Group, BMI Category, Diet Group, Sleep Group, Risk Quartile, Family History Diabetes Group, HbA1c Group, Physical Activity Group, Hypertension History Group, Fasting Glucose Group, Postprandial Glucose Group, Hypertension History Group** etc.

Analytical Segmentation:

Group	Categories
Age Group	Under 30, 30-44, 45-59, 60+
BMI Group	Underweight, Normal, Overweight, Obese
Diet Group	Poor, Moderate, Healthy
Sleep Group	Short, Moderate, Long
Risk Quartile	Q1, Q2, Q3, Q4
Physical Activity Group	Low, Medium, High
Fasting Glucose	Normal, Prediabetes Range, Diabetes Range
HbA1c Group	<5.7, 5.7-6.4, 6.5-7.4, 7.5, 8.4, >=8.5

7.4 Dashboard Design:

The final Power BI report consists of four dashboard pages:

01 - Executive Overview

Provides an overall view of diabetes prevalence, demographics, BMI, and risk-score patterns.

02 - Risk Factors

Examines lifestyle, medical history, and body-related factors associated with diabetes prevalence.

03 - Health & Biomarkers

Focuses on glucose, HbA1c, cholesterol, triglycerides, and blood-pressure indicators.

04 - High-Risk Population

Identifies high-risk, currently undiagnosed individuals and analyzes their demographic, BMI, and risk-score distribution.

7.5 Executive Overview:



Figure 5. Executive Overview Dashboard

Purpose:
Provides an executive-level summary of diabetes prevalence and major population patterns.

Key Finding:
Diabetes prevalence varies across age, BMI, gender, and risk-score segments.

7.6 Risk Factors:

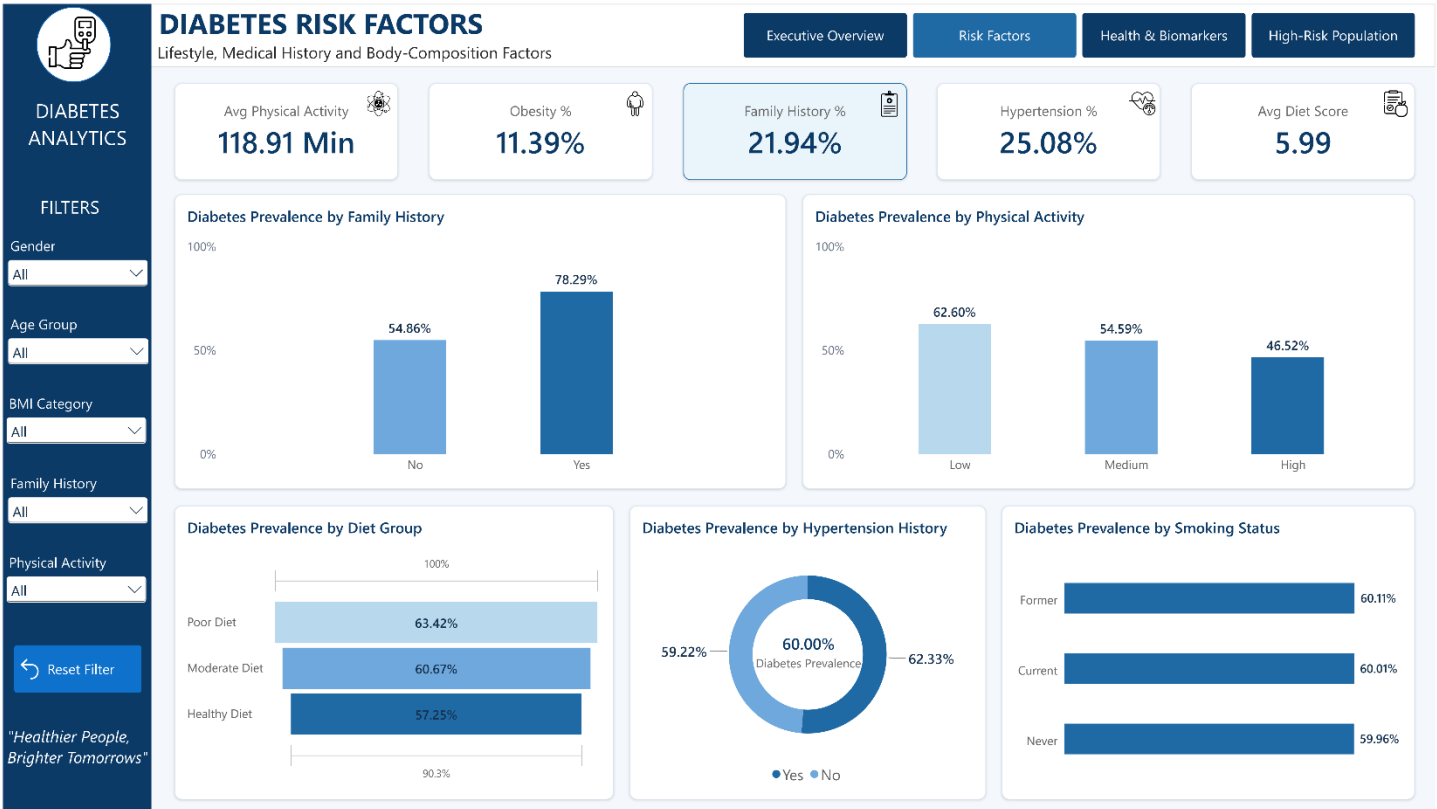


Figure 6. Risk Factors Dashboard

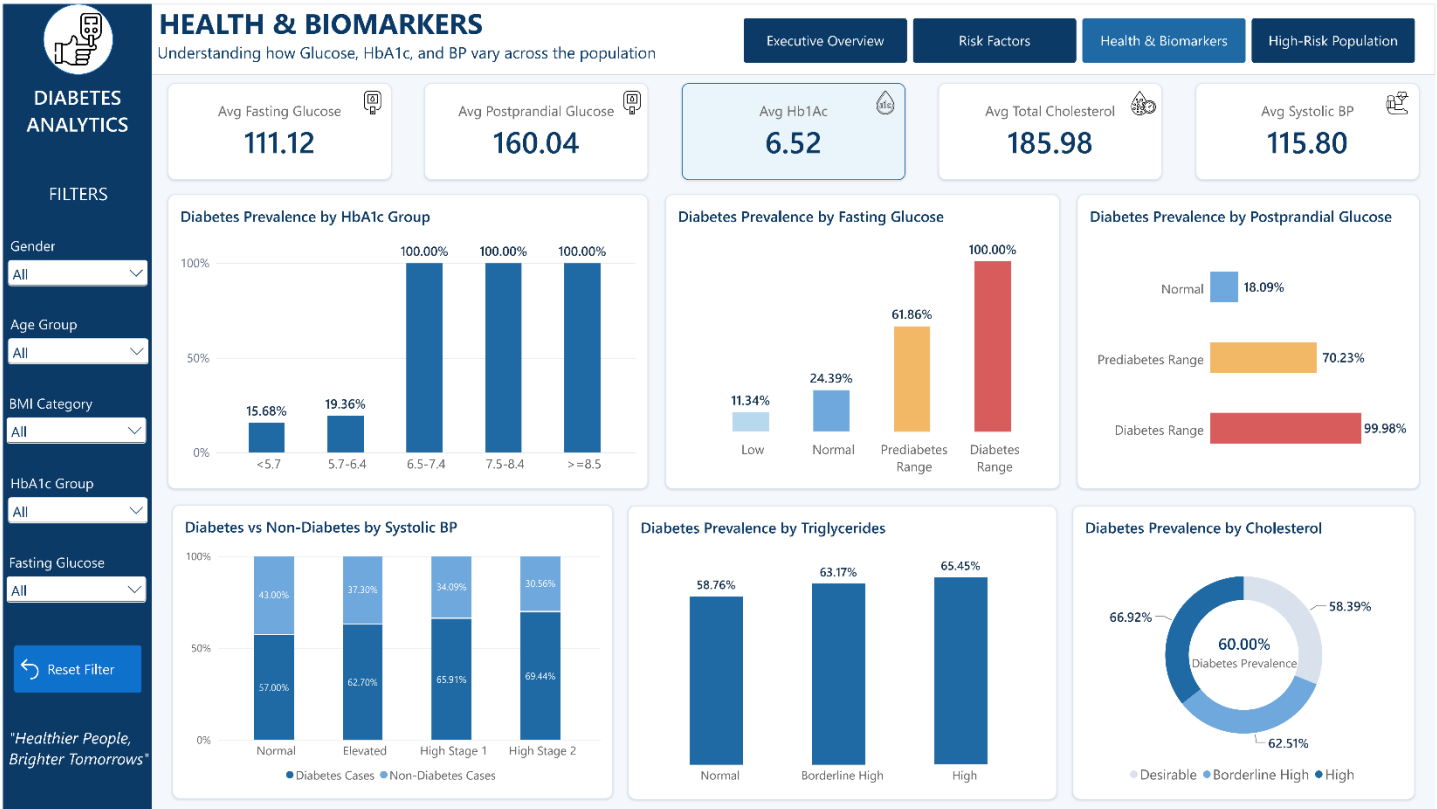
Purpose:

Examines lifestyle, medical-history, and body-related factors associated with diabetes prevalence.

Key Finding:

Family history, physical activity, BMI, and diet quality showed notable differences in diabetes prevalence.

7.7 Health & Biomarkers:



7.8 High-Risk Population:

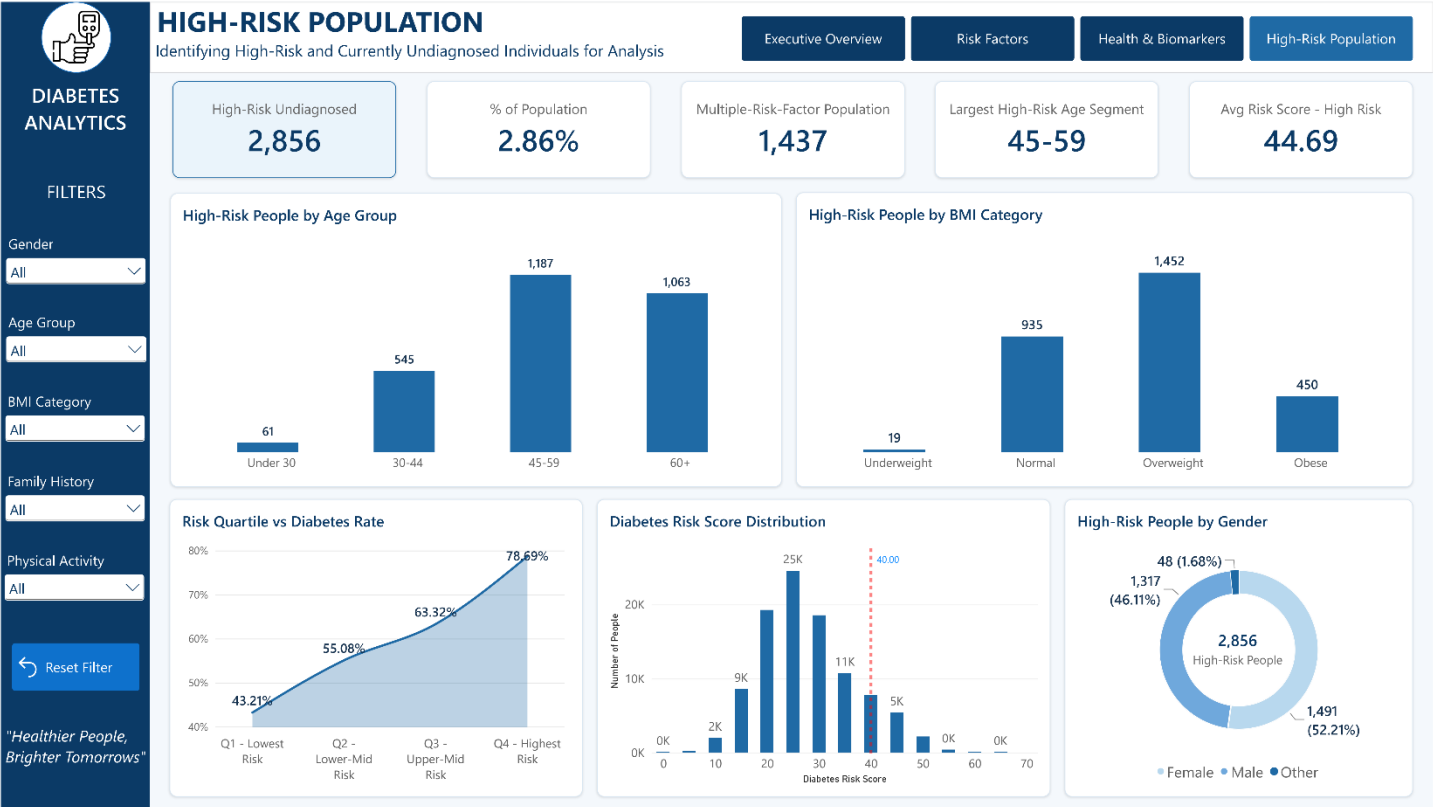


Figure 8. High-Risk Population Dashboard

Purpose:

Identifies and segments individuals with higher diabetes risk who are currently undiagnosed.

Key Finding:

The analysis identified **2,856 high-risk, currently undiagnosed individuals**, representing **2.86% of the total population**.

8. Key Findings

1. Overall Diabetes Prevalence

The dataset contains **100,000 individuals**, of whom **59,998 were diagnosed with diabetes**, resulting in an overall diabetes prevalence of **60.00%**.

2. Family History Shows a Strong Association

Diabetes prevalence was **78.29%** among individuals with a recorded family history of diabetes, compared with **54.86%** among those without a recorded family history.

Difference: 23.43 percentage points.

This was one of the strongest differences observed among the analyzed risk factors.

3. Physical Activity Shows a Clear Difference

Diabetes prevalence was **62.60%** in the low physical-activity group compared with **46.52%** in the high physical-activity group.

Difference: 16.08 percentage points.

4. BMI Is Associated with Higher Diabetes Prevalence

The obese group had a diabetes prevalence of **68.28%**, compared with **55.96%** among the normal-BMI group.

Difference: 12.32 percentage points.

5. Glucose and HbA1c Show Strong Differences

Diabetes prevalence increased substantially across glucose categories. The fasting-glucose analysis ranged from **11.34% in the low group to 100% in the diabetes-range group**.

HbA1c categories also showed substantial differences, with the diabetes-range group showing **100% diabetes prevalence**.

These biomarkers showed some of the strongest associations with diagnosed diabetes in the dataset.

6. Higher Risk Scores Are Associated with Higher Diabetes Prevalence

Diabetes prevalence increased across risk-score quartiles:

Risk Quartile	Diabetes Prevalence
Q1 – Lowest Risk	43.21%
Q2 – Lowest-Mid Risk	55.08%
Q3 – Upper-Mid Risk	63.32%
Q4 – Highest Risk	78.69%

The difference between Q1 and Q4 was **35.48 percentage points**.

7. High-Risk Undiagnosed Population

Using the project definition of **risk score ≥40 and diagnosed diabetes = 0**, the analysis identified **2,856 high-risk, currently undiagnosed individuals**, representing **2.86% of the total population**.

The largest high-risk age segment was **45–59 years**, with **1,187 individuals**.

9. Recommendations

1. Prioritize High-Risk Population Segments

Use the diabetes risk score to identify and prioritize individuals with higher risk, particularly those who are currently undiagnosed and meet multiple risk-factor criteria.

2. Focus on Preventive Monitoring

Individuals showing combinations of elevated risk score, glucose, HbA1c, and BMI could be prioritized for **further assessment and preventive monitoring**.

3. Promote Healthy Lifestyle Patterns

The observed differences across physical activity, diet quality, and BMI suggest that population-level wellness programs could focus on **increasing physical activity, improving diet quality, and maintaining healthy body weight**.

4. Incorporate Family History Into Risk Segmentation

Family history showed a substantial difference in diabetes prevalence. It can therefore be considered as an additional variable when developing **population-level risk segmentation and monitoring strategies**.

5. Monitor Key Biomarkers

Glucose and HbA1c showed particularly strong associations with diagnosed diabetes. These indicators should receive greater attention when analyzing and monitoring diabetes-related risk patterns.

6. Target Older and Higher-Risk Segments

Older age groups and higher risk-score segments showed higher diabetes prevalence. Analytics-based monitoring can therefore prioritize these segments for further investigation and preventive programs.

7. Use Dashboards for Ongoing Monitoring

The Power BI dashboard can be used to monitor diabetes prevalence, risk factors, biomarkers, and high-risk population segments across different demographic and health categories.

Important Note: These recommendations are based on patterns observed in the dataset and are intended for analytical and population-level decision support. They should not be interpreted as individual medical advice or clinical recommendations.

10. Conclusions

This project demonstrates an end-to-end data analytics approach to understanding diabetes prevalence and associated health and lifestyle patterns within a dataset of **100,000 individuals**.

The analysis combined **Python, PostgreSQL, and Power BI** to transform raw healthcare data into meaningful insights. Python was used for data cleaning, validation, and exploratory data analysis; PostgreSQL was used for business-focused and advanced analytical analysis; and Power BI was used for data modeling, DAX calculations, interactive visualization, and dashboard development.

The analysis identified notable differences in diabetes prevalence across **family history, physical activity, BMI, glucose levels, HbA1c, age groups, and risk-score segments**. Risk-score segmentation also helped identify a population of **2,856 high-risk, currently undiagnosed individuals** based on the project's analytical criteria.

The final four-page Power BI dashboard provides an interactive view of the overall population, risk factors, health and biomarkers, and high-risk population segments.

Overall, the project demonstrates the ability to take a **raw dataset through the complete analytics lifecycle — from data preparation and exploration to SQL analysis, visualization, insights, and recommendations**.

Note: The findings represent associations observed within the dataset and should not be interpreted as clinical diagnoses or causal relationship

11. Limitations

1. Dataset Representativeness

The analysis is based on the available dataset and may not represent the characteristics of the broader population. Therefore, the findings should be interpreted within the context of this dataset.

2. Observational Analysis

The analysis identifies **associations and patterns**, but it does not establish causal relationships between risk factors and diabetes.

3. Analytical Categories

Some categories, such as **diet groups, sleep groups, risk quartiles, and other analytical segments**, were created for the purpose of this analysis. These categories should not be interpreted as standardized clinical classifications unless supported by an appropriate clinical methodology.

4. High-Risk Population Definition

The high-risk undiagnosed population was identified using the project's analytical criteria, including a **diabetes risk score ≥ 40 and diagnosed diabetes = 0**. Being classified as high-risk does **not** mean that an individual has diabetes.

5. Clinical Interpretation

The project is designed for **data analytics and population-level insight**, not individual medical diagnosis or treatment recommendations. Clinical decisions require appropriate medical assessment and validated clinical guidelines.

6. Data Availability

The analysis is limited to the variables available in the dataset. Additional factors that may influence diabetes risk may not be captured.

Closing Note

Overall, these limitations do not prevent the dataset from being useful for analytical exploration. They define the appropriate scope within which the findings and recommendations should be interpreted.